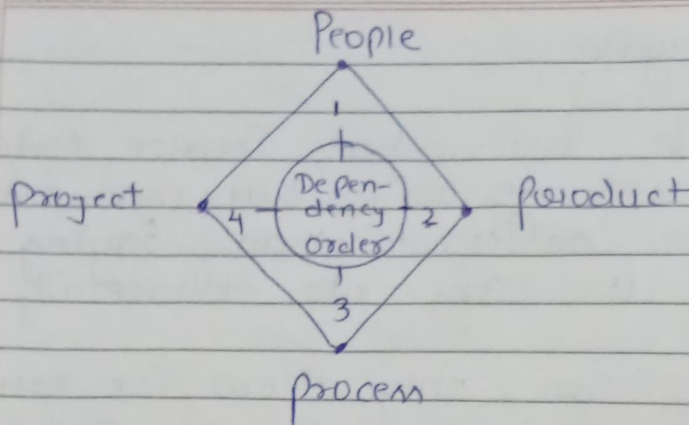


Role of Management in SW Development

The management of SW development is heavily dependent on 4 factors:

1. people
2. product
3. process
4. project



slw development is a people Centric activity. Hence, Success of project is on the shoulders of people who are involved in the development.

The people

slw development requires good managers. The managers, who can understand the psychology of people and provide good leadership. A Good Manager cannot ensure the Success of the project, but can increase the probability of Success. The areas to be given priority are:

- ① proper Selection
- ② training
- ③ Compensation
- ④ Career development

⑤ work culture.

The product

Objectives and Scope of work should be defined clearly to understand the requirements so not to occur a problem.

it may help the Managers to Select a "best" approach within constraints imposed by delivery deadlines, budgetary restrictions, personnel availability, technical interfaces etc.

without well defined requirements, it may be impossible to define reasonable estimates of the cost development time & Schedule for the project.

The Process

it is the way in which we produce sw. it provides the framework from which a comprehensive plan for sw development can be established. if the process is weak, the end product will suffer.

There are Many life cycle Models & process improvements Models.

Good Write

depending On the type of project a Suitable model is to be Selected. Now-a-days CMM (Capability Maturity Model) has become almost a Std. for process framework.

The project

A proper planning is required to Monitor the status of development & to Control the Complexity.

Most of the projects are Coming late with Cost Overruns of More than 100%. In Order to manage a Successful project, we Must understand what Can go wrong & how to do it right.

We should define Concrete Requirements & freeze these Requirements.

S/w Surprises are always Risky & we should Minimise them.

We should have a planning Mechanism to Give warning before the Occurrence of any Surprise.

sw life cycle Models

- ① The Software development life cycle (SDLC) is a structured process that is used to design, develop and test good quality sw.
- ② it is a methodology that defines the entire procedure of sw development step by step.
- ③ The goal is to deliver high quality, maintainable sw that meet the user's requirements.
- ④ SDLC outlines the plan for each stage so that each stage of sw development Model can perform its task efficiently to deliver the sw at a low cost within a given time frame that meets user's requirements.
- ⑤ it is a process followed for sw building within a sw organization.
- ⑥ it consists of a precise plan that describes how to develop, maintain, replace and enhance specific sw.
- ⑦ The life cycle defines a method for improving the quality of sw and the all-around development process.

⇒ what is the Need for SDLC

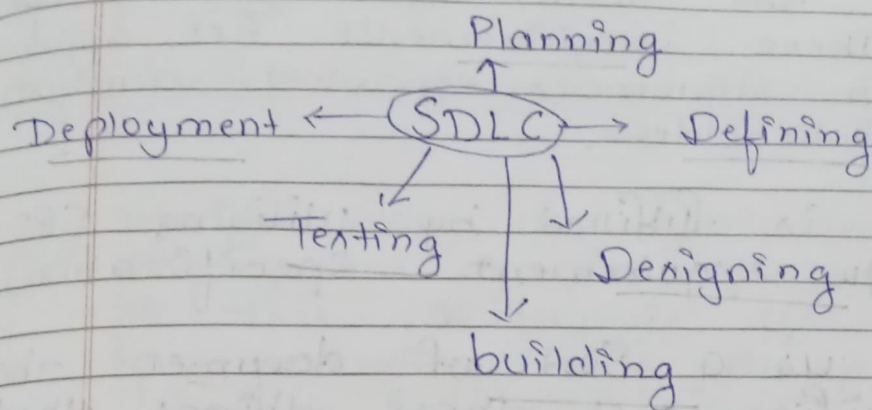
- ① it is a Method, approach or process that is followed by a software development organization while developing any software.
- ② These Models were introduced to follow a disciplined and systematic Method while designing software.
- ③ with SDLC, the process of software design is divided into small parts, which makes the problem more understandable & easier to solve.
- ④ SDLC comprises a Detailed description or step wise plan for designing, developing, testing & Maintaining the software.

Stages of SDLC Model

SDLC specifies the task to be performed at various stages by a software engineer or developer. It ensures that end product is able to meet the Customer's expectation and fits within the Overall budget.

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The SDLC Model involves 6 phases or stages while developing any s/w.



Stage 1: planning & requirement analysis.

planning is a crucial step in everything, just as in s/w development. In this same stage, requirement analysis is also performed by the developers of the organization. This is attained from customer inputs and sales department.

The information from this analysis forms the building blocks of basic project. The quality of project is a result of planning. Thus, in this stage, the basic project is designed with all the available information.

Stage 2: Defining Requirements

In this stage, all the requirements for the target SW are specified. These requirements get approval from customers, market analysts & stakeholders.

This is fulfilled by utilizing SRS (SW Requirement Specification).

This is a sort of document that specifies all those things that need to be defined & created during the entire project cycle.

Stage 3: Designing Architecture

SRS is a reference for SW Designers to come up with the best architecture for SW. Hence, with the requirements defined in SRS, Multiple designs for the product architecture are presented in the Design Document Specification (DDS).

This DDS is assessed by Market analysts & stakeholders.

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after evaluating all the possible factors, the most practical & logical design is chosen for development.

Stage 4 : Developing product

at this stage, the fundamental development of the product starts. for this, developers use a specific programming code as per the Design in the DDS.

Hence, it is important for the coders to follow the protocols set by the association.

Conventional programming tools like compilers, interpreters, debuggers etc. are also put into use at this stage.

Some popular languages like c/c++, python, java are put into use as per the slw regulations.

Stage 5 : product testing & Integration

after the Development of product, testing of slw is necessary to

ensure its smooth execution, although, minimal testing is conducted at every stage of SDLC.

Therefore, at this stage, all the probable flaws are tracked, fixed and re-tested. This ensures that the product confronts the quality requirements of SRS.

Documentation, Training and Support:

SW documentation is an essential part of the SW development life cycle. A well-written document acts as a tool and means to information repository necessary to know about SW processes, functions and maintenance.

Documentation also provide information about how to use the product.

Training is an attempt to improve the current or future employee performance by increasing an employee's ability to work through learning, usually by changing his attitude & developing his skills and understanding.

Stage 6 : Deployment & Maintenance of products.

after detailed testing, the Conclusive product is released in phases as per the Organization's Strategy. Then it is tested in a real Industrial environment.

it is Important to ensure its Smooth performance. if it performs well, the Organization Sends out the product as a whole.

After receiving beneficial feedback, the Company releases it as it is or with auxiliary Improvements to make it further helpful for the Customers.

However, this alone is not enough. Therefore, along with the deployment, the product's supervision.